



Artificial intelligence in the battle against disinformation and misinformation: a systematic review of challenges and approaches

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Abstract

In the rapidly evolving digital age, the proliferation of disinformation and misinformation poses significant challenges to societal trust and information integrity. Recognizing the urgency of addressing this issue, this systematic review endeavors to explore the role of artificial intelligence (AI) in combating the spread of false information. This study aims to provide a comprehensive analysis of how AI technologies have been utilized from 2014 to 2024 to detect, analyze, and mitigate the impact of misinformation across various platforms. This research utilized an exhaustive search across prominent databases such as ProQuest, IEEE Explore, Web of Science, and Scopus. Articles published within the specified timeframe were meticulously screened, resulting in the identification of 8103 studies. Through elimination of duplicates and screening based on title, abstract, and full-text review, we meticulously distilled this vast pool to 76 studies that met the study's eligibility criteria. Key findings from the review emphasize the advancements and challenges in AI applications for combating misinformation. These findings highlight AI's capacity to enhance information verification through sophisticated algorithms and natural language processing. They further emphasize the integration of human oversight and continual algorithm refinement emerges as pivotal in augmenting AI's effectiveness in discerning and countering misinformation. By fostering collaboration across sectors and leveraging the insights gleaned from this study, researchers can propel the development of ethical and effective AI solutions.

Keywords Artificial intelligence · Disinformation · Misinformation · Review

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1 Introduction

Disinformation refers to false information that is intentionally created and spread with the purpose of deceiving or manipulating people [1]. It is a deliberate attempt to mislead others by presenting false or misleading information as facts [2]. Disinformation can be used for various purposes, such as influencing public opinion, undermining trust in institutions, or creating social unrest [3]. Those who spread disinformation often have specific agendas or motives behind their actions, seeking to achieve certain goals by manipulating the beliefs or behaviors of others [2, 3].

Misinformation, on the other hand, is false information that is shared without the deliberate intent to deceive [4]. It can be the result of misunderstanding, misinterpretation, or simply sharing inaccurate information unknowingly [5]. Misinformation can spread rapidly, especially in today's digital age, where information can be easily shared through social media and other online platforms [2]. While misinformation may not be spread with malicious intent, it can still have harmful consequences by creating confusion, mistrust, and the dissemination of inaccurate information [6].

Both disinformation and misinformation pose significant challenges in today's information landscape, impacting trust, decision-making, social cohesion, public health, democracy, and cybersecurity [7–9]. Addressing these challenges requires a concerted effort to promote media literacy, critical thinking skills, fact-checking initiatives, and responsible information-sharing practices among individuals and organizations [10]. By raising awareness about the differences between disinformation and misinformation and empowering people to critically evaluate the information they encounter, we can work toward mitigating the harmful effects of false information in society [11].

Due to the increasing growth in artificial intelligence, one potentially useful solution to identify and prevent the spread of this false information could be the use of artificial intelligence [12]. Artificial intelligence can be interpreted as the capacity of a system to perform activities related to human intelligence, including learning and decision-making [13].

Artificial intelligence (AI) influences decision-making in a growing range of sectors and has the potential to increase the efficiency of detecting and identifying fake news promptly [14, 15]. Some researchers have suggested that artificial intelligence, with its reliance on computational capabilities rather than human actions, provides an unbiased response to misinformation [16]. With continued improvements in the accuracy and efficiency of AI systems, there is growing anticipation that machines can excel in areas where human decision-making has faltered, particularly in reducing personal biases [17].

The dissemination of false and misleading information poses a significant challenge in today's digital age [17–20]. As AI technology advances, it is imperative to understand how it can be leveraged to address this problem effectively. However, there is a need to examine the existing literature and identify the approaches and challenges associated with using AI in combating disinformation and misinformation comprehensively. The purpose of this study is to conduct a systematic review of the use of artificial intelligence (AI) in combating the spread of disinformation and misinformation.

2 Methodology

This study consists of a thorough review of the existing literature to explore the various aspects and indicators related to the use of AI to detect and combat disinformation and misinformation, particularly in the digital context. This study utilizes the PRISMA checklist and follows the recommended reporting guidelines for systematic reviews [21] in addressing the research questions that follow.

2.1 Research questions

The study is guided by the following research questions:

1. What are the different AI approaches and techniques used to detect and combat disinformation and misinformation?
2. What are the strengths and limitations of AI in addressing the challenges related to disinformation and misinformation?
3. How effective are AI-based systems in detecting and mitigating the spread of disinformation and misinformation?
4. What ethical considerations should be taken into account when using AI to combat disinformation and misinformation?

2.2 Inclusion and exclusion criteria

The study employed rigorous selection criteria to identify relevant articles from various databases. The researchers aimed to gather a diverse range of scholarly works without prioritizing any specific article types (e.g., review articles, original research, letters to the editor). This inclusive approach ensured that valuable insights were not overlooked.

The focus was specifically on papers that explored the intersection of AI and disinformation, particularly those discussing how AI can enhance the measurement and analysis of disinformation patterns. By applying these stringent criteria, researchers ensured that selected studies directly addressed the impact of AI techniques in disinformation analysis.

2.3 Databases and search method

A comprehensive search was conducted across multiple databases, including:

- ProQuest
- IEEE Explore
- Web of Science
- Scopus
- Google Scholar

The search targeted articles published between January 1, 2014, and March 31, 2024, ensuring coverage of recent literature relevant to the research goals. A combination of broad search terms was utilized with nested searches. For example, in Web of Science, search strings included terms like “AI” or “Artificial Intelligence” combined with “Disinformation” or “Misinformation,” along with their variations. Boolean operators were used to refine searches across databases.

Additional manual searches were conducted using Google Scholar and citation chaining to ensure comprehensive coverage. Further details on the search strategy are provided in Supplementary File 1.

2.4 Study selection

The initial step involved identifying articles that met predefined inclusion criteria. Researchers reviewed article titles and abstracts for relevance to the research questions. In cases of conflicting evaluations, articles were excluded based on consensus among team members. When uncertainty persisted, input from additional scholars was sought to ensure a robust selection process.

2.5 Quality assessment

The Critical Appraisal Skills Programme (CASP) Systematic Review tool [22] was employed for quality appraisal of selected studies. This tool assesses research articles based on three primary domains: research questions, design and methodology, and interpretation of results. Each domain is evaluated against specific criteria, contributing to a scoring system ranging from 1 to 10:

- Scores between 8 and 10: Highly relevant
- Scores above 6: Moderately relevant
- Scores below 6: Irrelevant

This systematic review involved collaborative efforts among three researchers, with discrepancies resolved through consensus with a fourth contributor.

2.6 Data extraction and synthesis

An interpretative approach was utilized for data extraction and synthesis from reviewed studies. Key findings were distilled through careful reading and interpretation to uncover overarching themes. Further details are provided in Supplementary File 2.

2.6.1 Approach

Studies focusing on AI's role in combating disinformation were identified across scholarly databases. Each selected study underwent scrutiny for methodologies, findings, and conclusions.

2.6.2 Synthesis

Through iterative readings and discussions among researchers, common themes emerged from the studies. Extracted data were synthesized to construct a comprehensive overview of challenges and approaches discussed in the literature, providing insights into strengths, limitations, and future directions for AI-driven solutions in combating misinformation.

3 Results

3.1 Study identification

Based on our database search strategy, which included ProQuest, IEEE Explore, Web of Science, Scopus, and Google Scholar, we initially identified a total of 8310 articles. The breakdown of relevant articles by database is as follows: 1905 from ProQuest, 2362 from IEEE Explore, 1920 from Scopus, and 1425 from Web of Science. Additionally, our manual search in Google Scholar yielded 160 articles.

After eliminating 3636 duplicate entries, we proceeded to screen the remaining 4665 articles. During this screening process, we excluded 3251 articles that were not aligned with the study objectives; 2761 articles belonged to other academic disciplines; and 490 were in languages other than English. This left us with 913 potentially eligible articles.

Further examination of titles and abstracts led to the exclusion of an additional 352 articles that did not meet our study design criteria (i.e., they addressed unrelated topics). We then assessed the full texts of the remaining 561 articles for inclusion. Following this thorough evaluation, we excluded another 488 articles, resulting in a final selection of 73 articles for our systematic review. Additionally, we discovered three relevant articles through citation chaining. Ultimately, our final summary includes a total of 76 articles (see Fig. 1).

3.2 Quality assessment result

The quality appraisal results of the studies conducted using the CASP tool revealed the following findings:

- Twenty studies obtained a score of 9, accounting for 26.31% of the total number of articles reviewed.
- There were twenty-five studies with a score of 8, representing 32.89% of the total.
- A total of sixteen studies received a score of 7, making up 21.05% of the reviewed articles.
- Additionally, fifteen studies obtained a quality score of 6, accounting for 19.73% of the total.

Overall, these findings indicate that the majority of the reviewed studies exhibited high level quality indicators. For more detailed information regarding the qualitative selection of the studies, please consult supplementary file 3.

3.3 AI approaches to detect disinformation and misinformation

There are several AI approaches and techniques used to detect and combat disinformation and misinformation. Natural language processing models can be utilized for the identification of improper language, grammar, and context of text, key indicators of possible incorrect or deceptive information. Several papers discussed machine learning or deep learning approaches to classify text as trustworthy information or misinformation, using actual pieces of misinformation in order to train the models. Sentiment analysis was also commonly mentioned, as disinformation is often characterized by highly polarizing content compared to reliable news sources (Table 1).

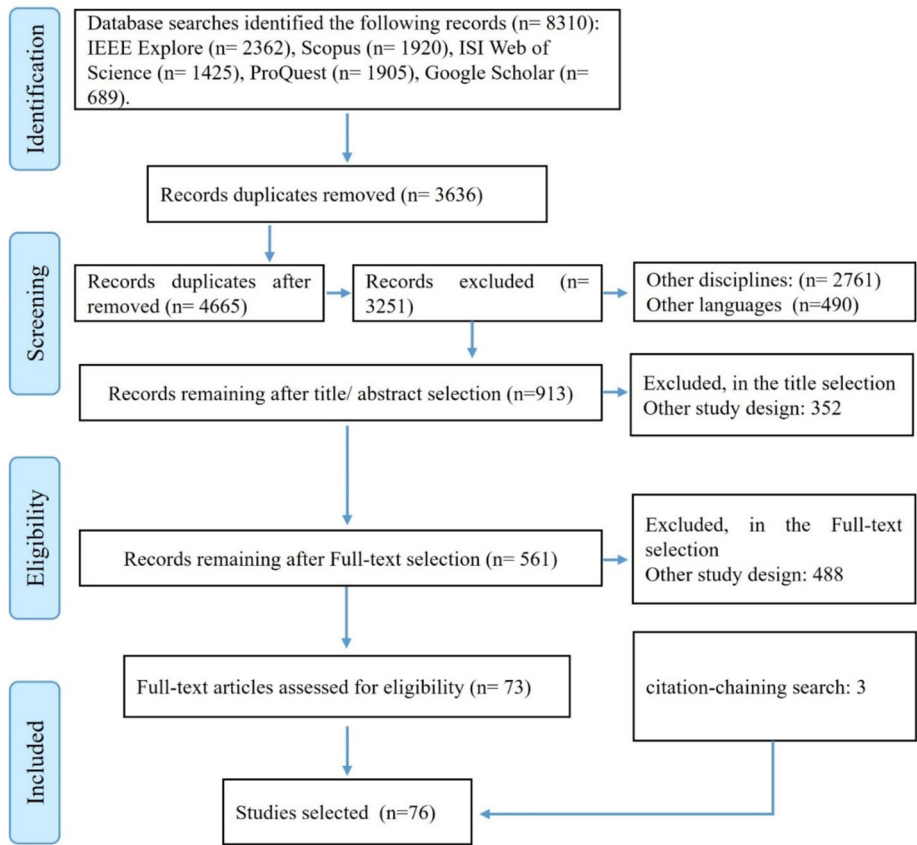


Fig. 1 Preferred reporting items for systematic reviews and meta-analysis (PRISMA) flow diagram of the selected studies

3.4 Strengths and limitations of AI in addressing misinformation and disinformation

AI has several strengths in addressing challenges related to disinformation and misinformation, but it also has some limitations, as shown in Tables 2 and 3. AI models are adaptable, meaning that they are able to adapt to new tactics being used to disseminate misinformation based on new training inputs. Similarly, they are scalable, able to adapt to ever-increasing sources of information online, while providing real-time monitoring of content as it is posted and shared. These models are also particularly adept at identifying patterns that may not be as obvious to a human observer, enabling increased accuracy and efficiency in the identification of false information.

Some of the major limitations of AI models stem from different biases and flaws that persist in many models today. AI models are only as adept as they are trained to be by their creator. Flawed training data, including incorrect data, biased data, or gaps in the data, can negatively impact the models and their ability to identify disinformation and misinformation. Often, the extent and nature of these biases are not even known to the human creators of the models. Additionally, over-reliance on these models could create a reduction in the critical

Table 1 AI approaches and techniques used to detect and combat disinformation and misinformation

AI approaches	Commentary	References
Deep learning	Deep learning techniques, such as neural networks, can be used to analyze complex patterns in large datasets. These techniques are particularly effective for detecting subtle forms of misinformation that may be difficult to identify using traditional methods	[23–26]
Machine learning	Machine learning algorithms can be trained on large datasets of known misinformation to identify patterns and features that are characteristic of false information. These algorithms can then be used to classify new information as either reliable or unreliable	[18, 27–32]
Natural language processing (NLP)	NLP techniques are used to analyze text data and identify patterns that indicate misinformation or disinformation. This can involve analyzing the language, grammar, and context of the text to detect inconsistencies or misleading information	[27, 28, 33–36]
Network analysis	Network analysis techniques can be used to analyze the spread of misinformation across social media platforms and other online networks. By identifying key nodes and patterns of dissemination, AI algorithms can help track the sources of misinformation and combat its spread	[2, 37–40]
Sentiment analysis	Sentiment analysis techniques can be used to analyze the emotions and attitudes expressed in text data. This can help identify misleading or biased content that is intended to manipulate readers' emotions	[41–46]

thinking and analysis skills that humans apply themselves in identifying potentially harmful information.

3.5 How effective are AI-based systems in detecting and mitigating the spread of disinformation and misinformation?

AI-based systems can be effective in detecting and mitigating the spread of disinformation and misinformation, but their effectiveness can vary depending on various factors, as shown in Tables 4 and 5. AI models must be highly efficient and adaptable to address evolving threats. As AI models evolve, so too will the creators of misinformation in order to avoid detection. As such, collaboration with humans who can help fine-tune the models, and the continued production of high-quality data for training and refinement are critical. These models cannot be expected to continue to operate effectively with no human involvement. Researchers expect that humans will work closely with AI models in order to ensure proper detection of disinformation and misinformation.

Table 2 Strengths of AI models in identifying disinformation and misinformation

AI strengths	Commentary	References
Adaptability	AI algorithms can be continuously updated and improved to keep up with evolving tactics used by those spreading misinformation	[14, 47, 48]
Automation	AI can automate the process of identifying and flagging potential misinformation, reducing the burden on human moderators and allowing for faster response times	[49–52]
Pattern recognition	AI can identify patterns and trends in data that may be indicative of misinformation, helping to detect false information more effectively	[29, 53, 54]
Real-time monitoring	AI can monitor online platforms in real-time to detect and respond to misinformation as it spreads, allowing for more timely interventions	[55–57]
Scalability	AI algorithms can analyze vast amounts of data quickly and efficiently, making it easier to detect and combat misinformation on a large scale	[58–61]

Table 3 Limitations of AI models in identifying disinformation and misinformation

AI limitations	Commentary	References
Bias	AI algorithms can be biased based on the data they are trained on, potentially leading to inaccurate or unfair detection of misinformation	[2, 62–64]
Context understanding	AI may struggle to understand complex nuances and context in language, leading to misinterpretations of information	[65–67]
Lack of transparency	Some AI algorithms are black boxes, meaning it is difficult to understand how they arrive at their decisions, which can be problematic for accountability	[12, 64, 68]
Over-reliance	There is a risk of over-reliance on AI in combating misinformation, which could lead to the neglect of human judgment and critical thinking	[69–72]

3.6 Ethical considerations when using AI to detect misinformation and disinformation

When using AI to combat disinformation and misinformation, it is crucial to consider ethical considerations and potential biases to ensure fair and effective outcomes. Because AI models are massive and complex, it can be easy for issues like data and algorithmic transparency to emerge, as well as issues with data privacy and accountability. Many of these issues persist in the popular AI models in use today. While many papers discuss these issues, there are very few that offer concrete approaches for handling them. This is an area that must be further addressed before the large scale deployment of AI models for the detection of disinformation and misinformation (Table 6).

Table 4 Factors Impacting the effectiveness of AI models in identifying disinformation and misinformation

AI effectiveness factors	Commentary	References
Algorithm accuracy	The accuracy of the AI algorithms used in detecting misinformation plays a crucial role in their effectiveness. Regular testing and validation are necessary to ensure the algorithms are reliable	[73–75]
Collaboration with human moderators	While AI can automate the detection process, human moderators are still essential for verifying and contextualizing information, enhancing the overall effectiveness of the system	[49–52]
Data quality	The effectiveness of AI systems in detecting misinformation relies on the quality and quantity of data they are trained on. High-quality, diverse training data can improve the accuracy of detection	[62, 76–80]
Real-time monitoring	AI systems that can monitor online platforms in real-time are more effective in detecting and responding to misinformation quickly, before it spreads widely	[12, 81]

Table 5 Challenges for AI models in successfully identifying disinformation and misinformation

AI challenges	Commentary	References
Adaptability	Misinformation tactics are constantly evolving, requiring AI systems to adapt quickly to new strategies and techniques	[43–45]
Bias	AI systems can be prone to bias based on the data they are trained on, potentially leading to inaccuracies in detecting misinformation	[2, 62–64]
Context understanding	AI systems may struggle to understand the complex nuances and context of language, leading to misinterpretations of information	[65–67]
Scale	With the vast amount of information shared online, AI systems may struggle to keep up with the sheer volume of data, impacting their effectiveness	[30–33]

4 Discussion

4.1 Summary of studies

Studies [27, 28, 33–36] focus on the use of natural language processing as a powerful tool for analyzing text data to detect inconsistencies and misleading information. These studies highlight how NLP algorithms examine language, grammar, and context to identify patterns indicative of misinformation. By leveraging these capabilities, NLP can effectively uncover misleading narratives and enhance the detection of false information in various textual formats [27, 28, 33–36].

Table 6 Ethical considerations for using AI models to detect disinformation and misinformation

AI ethical considerations	Commentary	References
Accountability	Clear mechanisms should be in place to hold AI systems and their developers accountable for their actions and decisions in combating misinformation	[12, 64, 68, 82–84]
Bias mitigation	Efforts should be made to identify and mitigate biases present in AI algorithms to ensure fair and accurate detection of misinformation	[20, 85–87]
Freedom of speech	Balancing the need to combat misinformation with the protection of freedom of speech is important to avoid censorship and promote open discourse	[19, 88–90]
Privacy	Respect for user privacy is essential when collecting and analyzing data to combat misinformation. Data should be handled securely and in compliance with privacy regulations	[83, 91, 92]
Transparency	AI systems used to combat misinformation should be transparent in their decision-making processes to ensure accountability and trustworthiness	[12, 64, 68]

Research referenced in studies [18, 27–32] explores the application of machine learning algorithms in classifying information as reliable or unreliable based on learned patterns from large datasets. These studies emphasize the effectiveness of sentiment analysis in detecting emotionally manipulative content and network analysis in tracking the spread of misinformation across online platforms. Together, these techniques provide a robust framework for identifying and mitigating disinformation.

Studies [23–26] investigate the effectiveness of deep learning techniques, particularly neural networks, in uncovering subtle forms of misinformation that may evade traditional detection methods. These advanced models are capable of learning complex patterns within large datasets, making them particularly valuable in identifying nuanced instances of false information that simpler models might miss.

The effectiveness of AI systems in combating disinformation is underscored by findings from studies [49–52, 58–61], which highlight their scalability, automation, pattern recognition capabilities, real-time monitoring, and adaptability to evolving tactics. However, these studies also raise important concerns regarding limitations such as algorithmic bias, challenges in understanding context, lack of transparency in decision-making processes, and the necessity for continuous refinement of algorithms to maintain effectiveness.

Ethical considerations surrounding the use of AI to combat disinformation are discussed in studies [12, 64, 68, 83, 91, 92]. These studies stress the importance of principles such as transparency, privacy protection, bias mitigation, and accountability in guiding the development and deployment of AI systems. They also address various biases—such as training data bias and algorithmic bias—that can compromise the fairness and accuracy of misinformation detection.

Studies [2, 93, 94] explore recent advancements in AI-based approaches for combating misinformation on social media platforms. They demonstrate that combining machine learning algorithms with NLP techniques has yielded promising results in accurately identifying false information through linguistic pattern analysis and contextual understanding.

Research findings from studies [62, 95, 96] illustrate the high accuracy of AI systems when combined with human oversight and continuous algorithm refinement over time. These longitudinal studies emphasize the adaptability and scalability required to effectively counter evolving misinformation tactics.

Lastly, studies [97–99] highlight research on human-AI collaboration in verifying news articles for misinformation. They showcase how the synergy between automated fact-checking tools and human judgment enhances reliability in verification outcomes, demonstrating the complementary roles that AI technologies and human expertise can play in combating misinformation.

4.2 Strengths of AI in combating disinformation

One of the primary advantages of AI in addressing disinformation lies in its diverse methodologies [35]. Natural language processing (NLP) is particularly notable for its ability to analyze text data to uncover inconsistencies and misleading information [35, 79]. By scrutinizing language, grammar, and context, NLP algorithms can identify patterns indicative of misinformation [35, 36, 79]. Similarly, machine learning algorithms utilize extensive datasets to classify information as credible or unreliable based on learned patterns [18, 27–32]. Techniques such as sentiment analysis can detect emotionally charged language commonly found in manipulative content, while network analysis tracks the dissemination of misinformation across digital platforms [41–46]. Advanced deep learning techniques, including neural networks, excel at recognizing subtle forms of misinformation that may evade traditional detection methods [23–26].

The effectiveness of AI in combating disinformation is further highlighted by its scalability, automation, pattern recognition capabilities, real-time monitoring, and adaptability to evolving tactics [62, 73–80]. However, several limitations must be considered. These include biases inherent in AI algorithms, challenges related to contextual understanding, lack of transparency in decision-making processes, over-reliance on AI solutions, and the necessity for continuous refinement of algorithms [2, 62–64].

4.3 Challenges and ethical considerations

The deployment of AI systems to combat disinformation raises significant ethical considerations. Key principles such as transparency, privacy, freedom of speech, bias mitigation, and accountability must guide the development and implementation of these technologies [64, 68, 85]. Various biases—such as training data bias, algorithmic bias, confirmation bias, cultural bias, and feedback loop bias—can adversely affect the fairness and accuracy of AI-driven misinformation detection [2, 62–64].

Addressing these biases is crucial for maintaining ethical standards in combating disinformation. Strategies may include employing diverse training datasets, conducting regular bias assessments, implementing human oversight mechanisms, utilizing explainable AI techniques, and engaging in ongoing monitoring [16, 100, 101].

Recent studies have explored numerous AI-based approaches for addressing misinformation on social media platforms [2, 93, 94]. Machine learning algorithms combined with NLP

techniques have shown promising results in accurately identifying false information through linguistic pattern analysis and contextual understanding [2, 93, 94].

Moreover, ethical frameworks have been examined in relation to leveraging AI for this purpose. Research highlights the importance of transparency, privacy protection, bias mitigation, and accountability throughout the development and deployment phases. The necessity for human oversight is emphasized to ensure ethical decision-making remains integral to these efforts [69–72].

Investigations into biases within AI algorithms used for misinformation detection have revealed significant insights. Biases arising from training data selection, algorithm design choices, and cultural influences can compromise the fairness and accuracy of detection efforts. Recommendations include implementing diverse training datasets and conducting regular bias assessments to effectively mitigate these issues [16, 100, 101].

Longitudinal studies assessing the efficacy of AI systems in combating misinformation over time indicate high accuracy rates when combined with human oversight and continuous algorithm refinement. These studies stress the adaptability required to effectively counter evolving tactics employed by purveyors of misinformation [12, 64, 68].

Research on human-AI collaboration has also underscored the complementary roles that automated fact-checking tools can play alongside human expertise in verifying news articles for misinformation. This synergy enhances reliability in verification outcomes.

4.4 Critical analysis

Natural language processing (NLP) is recognized for its ability to analyze text data and detect inconsistencies indicative of misinformation. Research indicates that NLP can achieve high accuracy in classifying news articles. For example, a study by Suhaib Kh et al. (2023) demonstrated that NLP techniques could classify news articles with over 90% accuracy [44]. However, challenges remain regarding context understanding; NLP models often struggle with sarcasm and idiomatic expressions, which can lead to misclassification [7, 28, 33–36]. This limitation highlights the need for ongoing advancements in contextual analysis within NLP frameworks.

Machine learning algorithms are effective in classifying information based on learned patterns from large datasets. Studies have shown that these algorithms can predict the spread of false information with significant accuracy [18, 27–32]. For instance, Vosoughi et al. [102] found that machine learning models could accurately forecast the dissemination of fake news on social media platforms. However, biases inherent in training datasets can compromise these models' effectiveness, perpetuating societal inequalities [2]. This concern emphasizes the necessity for diverse and representative datasets to ensure fairness in AI applications.

Sentiment analysis has emerged as a valuable tool for detecting emotionally charged content associated with misinformation. Research by Kai et al. (2017) illustrated how sentiment analysis could enhance misinformation detection by flagging content that evokes strong emotional responses. The study found that negative sentiment was more prevalent in user comments on fake news compared to real news [103]. However, sentiment analysis techniques can misinterpret nuanced expressions or fail to capture the subtleties of human emotion, indicating a need for further refinement [41].

Deep learning techniques, especially neural networks, have shown promise in uncovering subtle forms of misinformation that might evade traditional detection methods. Kai et al. (2017) demonstrated that deep learning models outperformed conventional classifiers in identifying fake news articles. Nonetheless, these models require substantial labeled training

data, which can be a limitation in rapidly changing information environments [103]. Ensuring comprehensive and representative training datasets is critical for maintaining the effectiveness of deep learning approaches.

Ethical considerations are paramount when deploying AI technologies to combat disinformation. Studies emphasize the importance of transparency, privacy protection, and accountability in AI systems [12, 64, 68, 82–84]. For instance, Jobin et al. [104] highlighted that without explicit ethical guidelines, organizations risk creating systems that exacerbate biases rather than mitigate them. This underscores the need for frameworks that guide ethical AI development.

The integration of human oversight into AI systems is essential for ensuring long-term effectiveness in combating misinformation. Research supports this notion; studies have shown that combining human judgment with AI tools significantly improves the accuracy of misinformation detection efforts [105]. This synergy between human expertise and AI technologies is crucial for addressing the complexities associated with disinformation.

5 Limitations

This study is not without limitations. Despite the extensive search strategy employed in this research, there remains the possibility of publication bias, as well as the limitation of focusing primarily on English-language publications. The temporal scope of 2014 to March 2024 may not capture the very latest advancements in AI technologies combating misinformation in the intervening months since the end of data collection. Furthermore, the inherent variability in database coverage and search methodologies may have influenced the comprehensiveness of our findings. Methodological differences among the included studies and the interpretative nature of our synthesis may introduce subjective biases into our review.

Nevertheless, this review provides valuable insights into the current landscape of AI applications in addressing misinformation challenges. By identifying key trends, ethical considerations, biases, and effective strategies, we contribute to the ongoing discourse on leveraging AI to combat misinformation effectively. Future research endeavors should aim to mitigate these limitations by expanding search criteria, including broader databases, and conducting more nuanced analyses of methodological rigor and contextual factors.

6 Conclusions

This systematic review underscores the significant potential of artificial intelligence (AI) as a powerful tool in the fight against disinformation and misinformation. By employing advanced techniques such as natural language processing (NLP), machine learning, sentiment analysis, network analysis, and deep learning, AI systems can effectively detect and mitigate false information on a large scale. However, to fully realize the capabilities of AI in this domain, it is essential to confront several challenges, including algorithmic bias, difficulties in context understanding, scalability issues, and ethical considerations.

To enhance the effectiveness of AI in combating misinformation, we recommend several key actions. First, future research should focus on identifying and mitigating biases in AI algorithms. This can be achieved by using diverse training datasets and implementing regular bias assessments to ensure fairness and accuracy in misinformation detection. Addressing

these biases is crucial for maintaining the integrity of AI systems and ensuring they serve all segments of society equitably.

Second, researchers should explore methods to improve AI's contextual understanding. Enhancing the ability of AI systems to grasp nuances in language and cultural factors will significantly increase their effectiveness in detecting misinformation. This could involve integrating more sophisticated NLP techniques that account for various situational influences on communication.

Additionally, investigating scalable solutions that can adapt to the rapidly evolving landscape of misinformation is crucial. Future research should focus on developing algorithms capable of efficiently processing large volumes of data from diverse sources in real-time. This adaptability will be key to keeping pace with the dynamic nature of information dissemination.

Incorporating human oversight into AI systems is another vital recommendation. The integration of human moderators can ensure ethical decision-making and contextual relevance in AI outputs. Future studies should examine effective models for collaboration between AI tools and human experts to improve the accuracy of misinformation detection.

Moreover, it is imperative that stakeholders prioritize ethical considerations in the development and deployment of AI technologies. Research should focus on establishing frameworks that promote transparency, accountability, and respect for privacy while combating disinformation. Ethical development practices will help build public trust in AI systems.

Finally, conducting longitudinal studies will provide valuable insights into the long-term effectiveness of AI systems in combating misinformation. These studies will allow researchers to assess how well these systems adapt over time and their performance in various contexts.

In an increasingly complex information landscape, the role of AI in addressing disinformation is set to become even more critical. While these technologies offer promising advancements in tackling one of the most pressing challenges of our time, it is essential that they are developed responsibly and ethically to avoid exacerbating existing problems. By understanding the challenges and strategies outlined in this review, stakeholders from various sectors can collaborate to create robust AI solutions that protect the integrity of information and foster a more trustworthy digital environment for all.

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